RAHEE

Week 3 Project Progress Report

Project	RAHEE, an online digital version of a travel board game
Team	Team 31 (5 members)
Report period	Week 3 (1 October - 8 October 2026)

Current stage

The multiplayer game is now feature-complete for the core rules, and this week was about the POC's two follow-up requests from the previous meeting and about hardening the server. Players can pick a destination and see the shortest route, invited players choose where they get off a road trip, and game rooms are saved to a database so a server restart no longer loses a game. At the weekly meeting on 7 October, the POC and CCL faculty member raised three bugs and two new requirements. We reproduced two of the bugs against the current board data and traced the third to the dealing code. Fixing these bugs is the priority for next week. The game is still not deployed.

1. Work Completed This Week

Items 1 and 2 below were requested by the POC in the Week 2 meeting. Items 3 and 4 are infrastructure work that followed from our Week 2 problem list.

1. **Destination-based movement:** A player picks a destination city instead of clicking one adjacent city at a time. The rules engine runs a shortest-path search (Dijkstra, weighted by steps) within the played card's allowance and shows the route. If the city cannot be reached on that card, the game shows a message saying so. The server re-checks reachability before it accepts the move.
2. **Rider-chosen drop-off on road trips:** An invited player now answers an invitation with one of three replies: accept and name the city where they get off (any city reachable on the proposed card), counter with a different destination, or decline. The driver then confirms or cancels. A rider is dropped automatically when the trip reaches their chosen stop.
3. **Server-side persistence:** Each room (seats and full game state) is written to a MySQL-compatible database before the new state is sent to any player. On start-up the server reloads every game still in progress. This closes the Week 2 problem that a server restart lost every game.
4. **Deployment groundwork:** The project is split into three independently deployable parts: a static frontend, the Node backend and the database. Database settings come from environment variables, the frontend's server address is set in one place, and the README has step-by-step deployment instructions. Nothing has been put online yet.
5. **Regression check:** The 80-game random simulation was re-run after these changes. All checks passed.

Status of the four items planned in Week 2

#	Item (from the Week 2 POC discussion)	Status
1	Shortest path to a chosen destination, with an out-of-reach message	Completed (see bug 2 below)
2	A way for players to discuss a road trip, separate from a general chat box	Not started. Only the counter-destination reply exists.
3	Invited rider chooses where they get off	Completed
4	Deploy so the game works outside the local network	Groundwork done, not yet online

2. Bugs Identified

These bugs were raised in the 7 October meeting with the POC and the design faculty member. We then checked each one ourselves. Priority is our own assessment.

ID	Bug	What we found	Status	Priority
B1	The route from Sawai Madhopur to Madikeri does not pass through the Gwalior node.	Reproduced. The search returns Sawai Madhopur, Bhopal, Lucknow, a flight to Bengaluru, then Madikeri (4 steps). The all-road alternative (8 steps) goes through Bhopal, Aurangabad and Hyderabad, and it offsets Gwalior.	Open	Medium
B2	A flight is offered as the shortest path even when a road path exists.	Reproduced. The search ranks routes by step count only, and a flight is one step, so it beats any longer road route. For example, New Delhi to Mumbai is returned as a single flight, while the road route takes 6 steps.	Open	High
B3	Dealing: after the guaranteed 10-valued cards are handed out, the leftover 10-valued cards are shuffled back into the deck. They should be set aside.	Confirmed by reading the dealing code. Only the guaranteed 10s are taken out. The rest of the deck, including the other 10-valued cards, is shuffled and dealt. This changes the cards players receive, and the rules file now says the remainder is set aside.	Open	High

Known issues carried over

- A player whose connection drops cannot rejoin their seat. The game state is safe in the database, but there is no way to reattach to it.

- Undo only reverses the last single movement step, not a whole turn action.
- Every route uses a placeholder of one step because the exact connection data has not been supplied. This also affects the route search results above.
- The browser walkthrough and screenshot tools still do not work with the new lobby flow, and the multiplayer check is a one-off script that is not kept in the project.

3. New Requirements from the Meeting

1. **Destinations of other players.** Each other player's destinations should be visible in a click-to-open panel on the left. The POC will give us the lists.
2. **New artwork.** PNG files of stamps, character cards, the map with nodes and a one-page rules sheet are expected early next week. Stamps will appear on hover at the nodes, and the maps are for the clickable left panel. We may need to talk to the design faculty member about how these are used.

4. Plan for Next Week

1. Fix B2: make the route search prefer road paths and offer a flight only when no road path exists within the card's allowance.
2. Fix B3: set aside the leftover 10-valued cards in the deal and add a test for it for every player count.
3. Fix B1: re-check the Bhopal-Lucknow link and the other routes around Gwalior against the printed board, then confirm Sawai Madhopur to Madikeri goes through Gwalior.
4. Add regression tests for B1, B2 and B3 so they cannot return.
5. Build the click-to-open panel showing other players' destinations.
6. Add the new artwork when it arrives (hover stamps, character cards, node map) and schedule a conversation with the design faculty member.
7. Build the private road-trip discussion between two players (Week 2 item 2).
8. Add reconnection so a dropped player can rejoin their seat.

5. Time Spent by Each Team Member, last Week

Hours are divided equally across the team for the previous week's work, covering the task areas below.

Name	Task(s)	Approx. Hours/week
Tanuj Sharma	Destination-based movement and shortest-path display (engine.js, main.js)	3
Sai Gawali	Road-trip invitation replies and rider-chosen drop-off (engine.js, handlers.js)	3
Siddhesh Patil	Database persistence and room restore on restart (backend/db.js, rooms.js, schema.sql)	3
Mukesh Dewangan	Frontend and backend separation, configurable server address, deployment steps (config.js, README)	3

Name	Task(s)	Approx. Hours/week
Kshitij Kasodkar	Re-running the random-game tests, checking the new bugs against the board data, and updating the documentation	3

6. Last Meeting with the POC

Date of last meeting	7 October 2026 (Wednesday), 2:00 pm
Meeting cadence	Weekly, every Wednesday at 2:00 pm.
Mode	Offline
Others present	Prof. Parvathy Subramanian, CCL faculty [Visual Designer]. She will provide designs, aesthetics and ideas for the game.

Key discussion / action items

1. The path from Sawai Madhopur to Madikeri did not touch the Gwalior node (bug B1).
2. Destinations of each other player should be visible in a click-to-open panel on the left. The POC will give us the lists (requirement 1).
3. After the 10-valued cards are distributed, the remaining cards should be removed, not shuffled with the rest. This is now commented in the rules (bug B3).
4. If a walkable path is available, a flight should not be offered as the shortest path (bug B2).
5. PNG files of stamps, character cards, the map with nodes and a one-page rules sheet will be given early next week. Stamps will show on hover at nodes, and the maps are for the clickable left panel. If needed, the team should schedule a time to talk to Prof. Parvathy next week (requirement 2).

7. Date of the Current Build

Current build date	7 October 2026
Build / version identifier	Commit f4524ed, main branch

8. Test Results on the Current Build

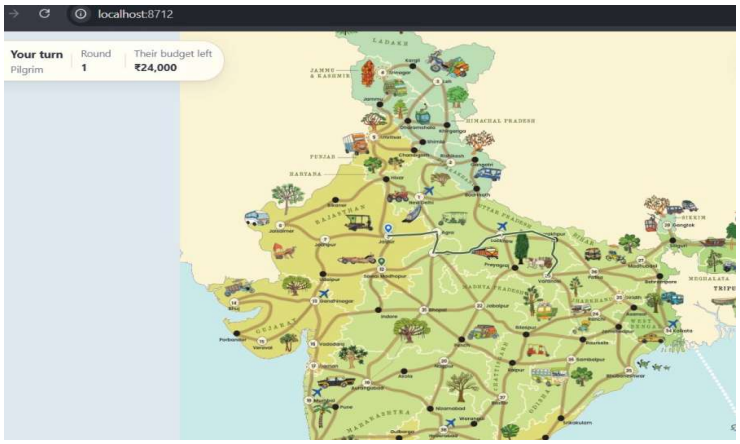
Tests are run against the rules engine with a script that plays a large number of complete random games and checks the rules after every one of them. The script was re-run on the current build.

Total tests	80 complete random games, each checked against every rule (deck composition, legal movement, scoring, and the two worked examples from the manual)
Unsuccessful tests	0
Successful tests	80

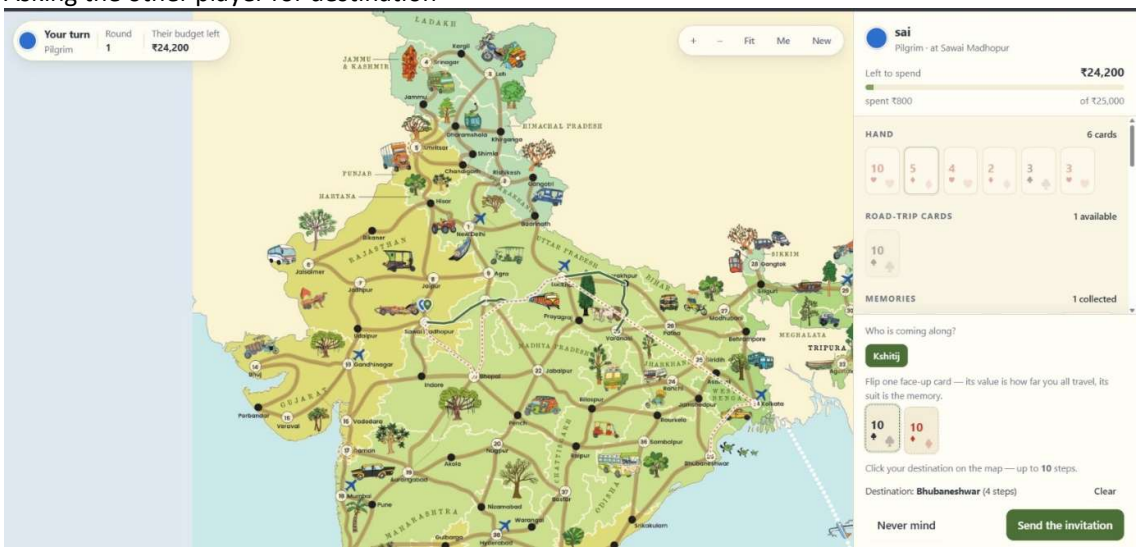
The passing result does not mean the three bugs above are absent. The test checks deck composition but not whether leftover 10-valued cards are set aside, and it does not check which route the search chooses. We reproduced B1 and B2 separately by running the search on the current board data and confirmed B3 by reading the dealing code. The destination search, the rider drop-off and database saving were checked by hand and have no automated tests yet.

Screenshots of the game

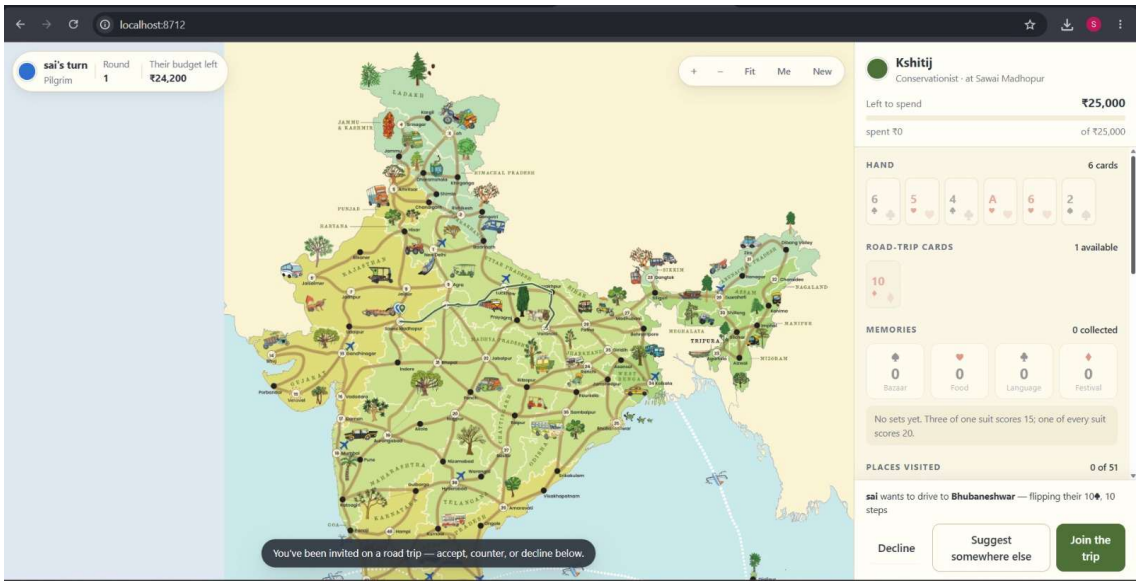
1. Destination picked with the shortest route shown



2. Asking the other player for destination



3. Invited player choosing to accept or to reroute



4. The Sawai Madhopur to Madikeri route showing bug B1

